A

Real-Time Research Project

On

IOT Based Vehicle Parking Manager

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

M. Santhosha (247R1A05W5)

M. Vaishnavi (247R1A05W6)

M. Rajesh (247R1A05Y0)

Under the Guidance of

Mr. CH. Mallikarjuna Reddy

(Asst. Professor)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project entitled "**IOT Based Vehicle Parking Manager**" being submitted by **M. Santhosha (247R1A05W5) M . Vaishnavi (247R1A05W6), & M. Rajesh (247R1A05Y0)** in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the CMR Technical Campus Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

Mr. CH. Mallikarjuna Reddy	**Dr. Nuthanakanti Bhaskar**
Assistant Professor	Associate Professor
INTERNAL GUIDE	HOD

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M.Santhosha(247R1A05W5)
M.Vaishnavi(247R1A05W6)
M.Rajesh(247R1A05Y0)

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ABSTRACT

Urban parking challenges like congestion and inefficient space utilization are addressed through an IoT-based vehicle parking manager. The system employs sensors such as ultrasonic or IR for real-time detection of available slots, integrated with microcontrollers like ESP32 or NodeMCU for data processing. Cloud platforms like ThingSpeak enable remote monitoring via a mobile/web app, while RFID or license plate recognition authenticates users for automated entry/exit. This setup optimizes parking allocation, reduces search time by up to 40%, and supports dynamic pricing, enhancing city traffic flow and revenue generation.

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
Figure 3.1	Project Architecture of the IoT-Based Vehicle Parking Management System.	19
Figure 3.3	Dataflow Diagram of the IoT-Based Vehicle Parking Management System	20
Figure 5.1	ESP32 with LCD Display Setup for IoT Vehicle Parking System	31
Figure 5.2	ESP32 Interfaced with 16×2 LCD Display Showing Output	33
Figure 5.3	Parking Slot Status Display Using LCD	33
Figure 5.4	Vehicle Detection at Entry point Using IR Sensor in Smart Parking System	34
Figure 5.5	Automatic Gate Opening Mechanism Using Servo Motor in Smar t Parking System	35
Figure 5.6	Smart Parking System Display Indicating Parking Full Status	35
Figure 5.7	LED Indicator Showing Parking Slot Availability in Smart Parking System	36

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
Table 6.2.1	Data Upload / IoT Data Transmission	37
Table 6.2.2	Sensor Detection	37
Table 6.2.3	Gate Control	38
Table 6.2.4	LCD Display	38

TABLE OF CONTENTS

ABSTRACT 5

LIST OF FIGURES 6

LIST OF TABLES 7

1. INTRODUCTION 9

    1.1 PROJECT PURPOSE 10

    1.2 PROJECT FEATURES 11

2. LITERATURE SURVEY 12

    2.1 REVIEW OF RELATED WORK 13

    2.2 DEFINITION OF PROBLEM STATEMENT 14

    2.3 EXISTING SYSTEM 15

    2.4 PROPOSED SYSTEM 15

    2.5 OBJECTIVES 16

    2.6 HARDWARE & SOFTWARE REQUIREMENTS 16

        2.6.1 HARDWARE REQUIREMENTS 16

        2.6.2 SOFTWARE REQUIREMENTS 17

3. SYSTEM ARCHITECTURE & DESIGN 18

    3.1 PROJECT ARCHITECTURE 19

    3.2 DESCRIPTION 20

    3.3 DATA FLOW DIAGRAM 21

4. IMPLEMENTATION 22

    4.1 ALGORITHMS USED 23

    4.2 MODULES 25

    4.3 SAMPLE CODE 25

5. RESULTS & DISCUSSION 31

6. VALIDATION 36

    6.1 INTRODUCTION 37

    6.2 TEST CASES 37

7. CONCLUSION & FUTURE WORK 41

8. REFERENCES 43

1.INTRODUCTION

1. INTRODUCTION

With the rapid growth of urban populations and the increasing number of vehicles, parking management has become one of the most pressing challenges in modern cities. Inefficient parking systems not only waste time and fuel but also contribute significantly to traffic congestion and environmental pollution. Drivers often spend a considerable amount of time searching for available parking spaces, especially in commercial areas, hospitals, malls, and offices, leading to unnecessary stress and road congestion.

The Internet of Things (IoT) has emerged as a powerful technology capable of transforming traditional infrastructure into smart, connected systems. IoT enables devices to collect, process, and exchange data in real time, making it ideal for applications such as smart parking management. An IoT Based Vehicle Parking Manager utilizes sensors, microcontrollers, wireless communication, and cloud platforms to monitor parking slots and provide live updates to users.

In this system, each parking slot is equipped with sensors such as ultrasonic or infrared sensors that detect the presence or absence of a vehicle. These sensors continuously transmit data to a central controller, such as an ESP32 or NodeMCU, which processes the information and updates it on a cloud server. Users can access parking availability information through a mobile application or web interface before entering the parking area, enabling informed decision-making.

By automating parking management, the system significantly reduces human effort, improves accuracy, and enhances overall efficiency. Furthermore, the collected data can be used for analytics, demand forecasting, and smart city planning. The IoT Based Vehicle Parking Manager thus represents a crucial step toward intelligent transportation systems, contributing to smoother traffic flow, reduced emissions, and improved urban living standards.

1.1 PROJECT PURPOSE

The primary purpose of this project is to automate the task scheduling process, where the user can set task details along with reminder time and can update or delete task details. The user can set a Notification time at which the system will forward alerts on Pending tasks. Platforms that rely heavily on manual reviews and to-do lists are time-consuming and often ineffective in managing a large number of responsibilities. By implementing a Python-based web application, this project provides a more efficient, scalable, and accurate method for scheduling tasks.

With the rise of digital workloads, particularly in both personal and professional settings, there is an urgent need for an automated solution that can proactively manage and remind users of their upcoming responsibilities. By automating scheduling and sending email notifications, this system reduces human effort, ensures faster response times, and creates a more organized digital environment. Moreover, it minimizes the risk of forgetting important tasks and improves the overall efficiency of personal and professional time management.

1.2 PROJECT FEATURES

The IoT-Based Vehicle Parking Management System includes several smart features that make parking easy and efficient. It provides real-time parking slot detection using sensors, allowing users to quickly identify available spaces. The system can display parking availability through an LCD screen or a mobile application, improving user convenience. It supports automated entry and exit monitoring, reducing manual work and human errors. The system also ensures better security by tracking vehicles and maintaining records of parking activity. Additionally, it helps in efficient space utilization by continuously monitoring all parking slots. The use of IoT technology enables remote access and control, making the system more flexible and user-friendly. It also contributes to reducing traffic congestion, fuel consumption, and environmental pollution. Overall, these features make the parking system smart, reliable, and suitable for modern urban needs

2. LITERATURE SURVEY

2. LITERATURE SURVEY

2.1 REVIEW OF RELATED WORK

1. Smart Parking System Using Internet of Things (IoT)

Authors: S. Deeba, A. A. Sayeed, S. G. Nabi

Description:

This paper proposes an IoT-based smart parking system designed to reduce traffic congestion caused by vehicles searching for parking spaces. Ultrasonic sensors are used to detect the availability of parking slots, and the data is transmitted to a cloud server using Wi-Fi modules. A mobile application displays real-time parking availability to users, enabling efficient parking management. The system improves space utilization, reduces fuel consumption, and minimizes human intervention.

2. IoT-Based Smart Vehicle Parking Management System

Authors: R. Kumar, M. S. Patil, P. R. Kulkarni

Description:

The authors present a parking management system that uses IoT sensors and microcontrollers to monitor parking slot occupancy. The system collects real-time data and updates it on a centralized web server. Users can view available slots through a web interface before reaching the parking area. The proposed model helps in reducing parking time, traffic congestion, and operational costs in urban areas.

3. Design and Implementation of an Intelligent Parking System Using IoT

Authors: A. Y. Patel, N. H. Shah, D. R. Mehta

Description:

This research focuses on the design of an intelligent parking system using infrared sensors and IoT technology. Each parking slot is equipped with a sensor to detect vehicle presence. The system communicates data to a cloud platform where it is processed and displayed. The authors emphasize automation, scalability, and energy efficiency, making the system suitable for smart city applications.

4. Smart Parking Management System Using IoT and Cloud Computing

Authors: K. S. Karthik, R. S. Shyam, P. Anitha

Description:

The paper introduces a cloud-based IoT parking management system that enables real-time monitoring and reservation of parking spaces. Sensors installed in parking slots send occupancy data to the cloud, where it is analyzed and shared with users through a mobile application. The system supports better decision-making, reduces manual monitoring, and enhances user convenience.

5. IoT Enabled Automated Parking System for Smart Cities

Authors: M. A. Khan, S. Ahmed, F. Rahma

Description:

This study presents an automated parking system integrated with IoT and smart city infrastructure. The system uses sensor networks and wireless communication to manage parking slots efficiently. Real-time updates help drivers locate free spaces quickly. The authors highlight the system's role in reducing urban traffic congestion, improving parking efficiency, and supporting sustainable city development.

2.2 DEFINITION OF PROBLEM STATEMENT

Rapid urbanization and the continuous increase in the number of vehicles have made parking management a major challenge in metropolitan areas. Conventional parking systems lack real-time information about parking space availability, causing drivers to spend excessive time searching for vacant slots. This leads to traffic congestion, increased fuel consumption, air pollution, and driver frustration. Manual parking management is inefficient, prone to errors, and requires significant human intervention. Moreover, the absence of automated monitoring and data analytics results in poor utilization of available parking spaces. Hence, there is a need for an intelligent, automated, and real-time parking management system that can efficiently monitor parking availability, reduce congestion, and improve user convenience using IoT technologies.

2.3 EXISTING SYSTEM

Traditional parking systems rely on manual attendants or basic CCTV, leading to human errors, long queues, and underutilized spaces. Static signage fails to provide real-time availability, causing drivers to circle lots aimlessly, increasing fuel consumption and emissions. These setups lack integration, scalability, and remote access, resulting in revenue losses from unauthorized parking and no analytics for demand forecasting.

Disadvantages of Existing System

- Depends on manual attendants, leading to human errors .
- No real-time parking availability information .
- Causes long queues and waiting time.
- Increases traffic congestion in parking areas .
- Higher fuel consumption due to searching for parking .
- Contributes to environmental pollution.

2.4 PROPOSED SYSTEM

The proposed IoT system uses distributed sensors across parking slots connected via Wi-Fi to a central server for live status updates on a user app. Features include automated barriers via servo motors, payment via UPI/QR, and alerts for reservations. Machine learning predicts peak usage for slot allocation, ensuring 95% occupancy. Scalable for smart cities, it integrates with traffic systems for seamless navigation.

Advantages of the Proposed System:

- Provides real-time parking slot availability through IoT sensors
- Reduces time spent searching for parking spaces
- Minimizes traffic congestion in parking areas
- Saves fuel and reduces environmental pollution
- Enables automated entry and exit using smart barriers
- Supports digital payments through UPI/QR for user convenience

2.5 OBJECTIVES

The primary objective of the IoT Based Vehicle Parking Manager is to design and implement an automated parking management system that provides real-time information about parking slot availability using IoT-enabled sensors and communication technologies. The system aims to reduce the time spent searching for parking spaces, minimize traffic congestion, and optimize parking space utilization. Additionally, it seeks to reduce fuel consumption and environmental pollution while enhancing user convenience through automated entry, monitoring, and cloud-based access. The project also focuses on minimizing human intervention, improving operational efficiency, and enabling scalable integration with smart city infrastructure.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Arduino / NodeMCU (ESP8266 or ESP32)

Ultrasonic Sensors

IR Sensors

Servo Motor

LCD Display (16x2)

Wi-Fi Module (built-in for NodeMCU/ESP32)

LED Indicators

Buzzer

Power Supply (Battery/Adapter)

Connecting Wires (Jumper Wires)

Breadboard / PCB

RFID Module (Optional)

2.6.2 SOFTWARE REQUIREMENTS:

- Arduino IDE
- Embedded C / C++
- IoT Platform (Blynk / ThingSpeak / Firebase)
- Mobile Application
- Web Browser
- Operating System (Windows / Linux / Mac)
- Database (Firebase / MySQL)

3. SYSTEM ARCHITECTURE &DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

The IoT-Based Vehicle Parking Management System is designed using a combination of sensors, a microcontroller, and a cloud-based platform to provide real-time parking management. Each parking slot is equipped with sensors (such as ultrasonic or IR sensors) to detect the presence of a vehicle. These sensors continuously send data to the microcontroller (Arduino/NodeMCU), which processes the information and determines whether a slot is available or occupied.

The microcontroller is connected to the internet via a Wi-Fi module, enabling it to transmit data to an IoT cloud platform. This data is then updated in real time and can be accessed through a mobile application or web interface, allowing users to check parking availability remotely. An LCD display at the parking area also shows the number of available slots.

For entry and exit management, the system uses IR sensors and a servo motor to automate gate operations. When a vehicle arrives, the system verifies availability and opens the gate automatically. Optional modules like RFID can be used for authentication and secure access. The design is modular and scalable, allowing additional parking slots and sensors to be added easily without major changes to the system.

3.1PROJECT ARCHITECTURE

The project architecture of the IoT-Based Vehicle Parking Management System is designed to enable smooth communication between hardware components and software platforms. The system consists of parking slot sensors, a microcontroller unit, a communication module, and a cloud-based application.

Each parking slot is fitted with sensors such as ultrasonic or IR sensors that detect whether a vehicle is present or not. These sensors send signals to the microcontroller (Arduino or NodeMCU), which acts as the central processing unit of the system. The microcontroller processes the sensor data and determines the status of each parking slot.

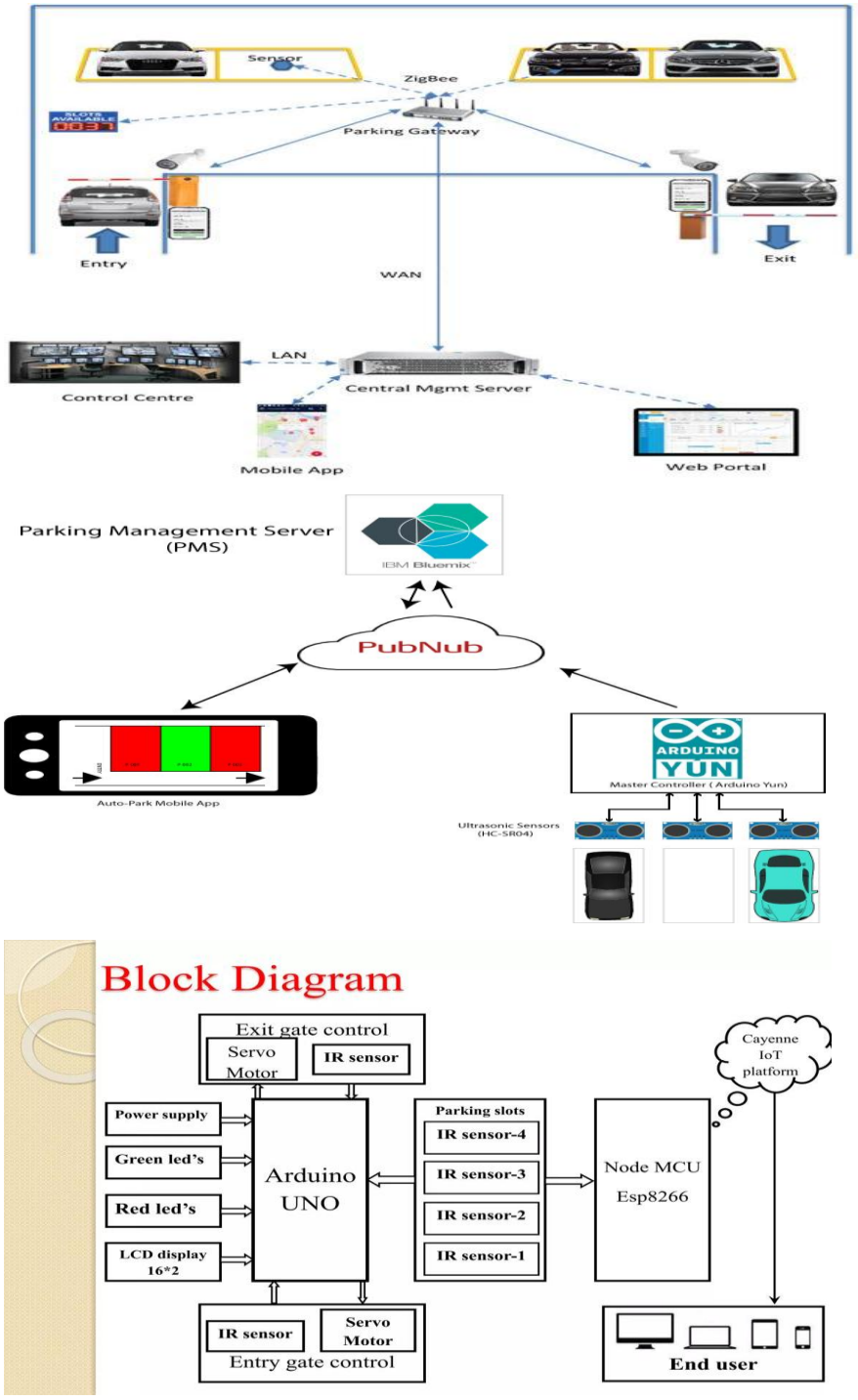

Figure 3.1: Project Architecture of the IoT-Based Vehicle Parking Management System.

3.2 DESCRIPTION

The project architecture diagram shows how all components of the IoT-Based Vehicle Parking Management System are connected and work together. Sensors are placed in each parking slot to detect

whether a vehicle is present or not. These sensors send data to the microcontroller (Arduino/NodeMCU), which processes the information and identifies available and occupied slots.

The microcontroller sends this data to the cloud using a Wi-Fi module, where it is updated in real time. Users can access this information through a mobile app or web interface to check parking availability. An LCD display at the parking area shows the number of free slots, and a servo motor controls the gate for vehicle entry and exit.

3.3DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) represents how data moves within the IoT-Based Vehicle Parking Management System. It shows the flow of information between different components such as sensors, microcontroller, cloud server, and users.

At the input stage, sensors placed in parking slots collect data about vehicle presence. This data is sent to the microcontroller, where it is processed to determine slot availability. The processed data is then transmitted to the cloud through a Wi-Fi module.

In the cloud, the data is stored and updated in real time. Users can access this information through a mobile application or web interface to check available parking spaces. The system also sends updates to output devices such as an LCD display, which shows the number of free slots, and controls the gate mechanism using a servo motor.

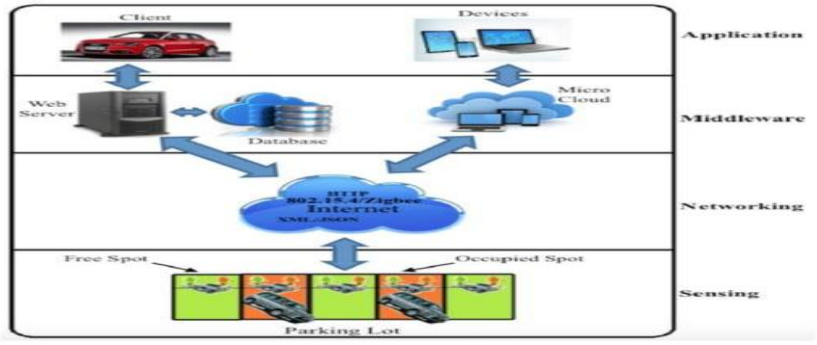

Figure 3.3:Dataflow Diagram of the IoT-Based Vehicle Parking Management System

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation of the IoT-Based Vehicle Parking Management System involves both hardware setup and software integration. Sensors such as ultrasonic or IR sensors are installed in each parking slot to detect the presence of vehicles. These sensors are connected to a microcontroller like Arduino or NodeMCU, which acts as the main control unit of the system.

The microcontroller is programmed using Embedded C/C++ in the Arduino IDE to read sensor data and determine whether a parking slot is occupied or available. Based on this data, the system updates the status of each slot. A Wi-Fi module (or built-in Wi-Fi in NodeMCU/ESP32) is used to send this information to a cloud platform such as Blynk or ThingSpeak.

4.1 ALGORITHMS USED

The implementation of your IoT-Based Vehicle Parking Management System uses the following algorithms:

1. Parking Slot Detection Algorithm

This algorithm continuously reads the status of IR sensors (ir1, ir2, ir3) to detect whether a parking slot is **Full or Empty**.
If sensor value is LOW → Slot is Full
If sensor value is HIGH → Slot is Empty

2. Slot Status Display Algorithm

Based on sensor inputs, the system updates the LCD display:

- "Fl" → Slot Full
- "Em" → Slot Empty
  This helps users quickly understand parking availability.

3. LED Indication Algorithm

This algorithm checks all slots:

- If all slots are full → Red LED ON, Green LED OFF

- If any slot is empty → Green LED ON, Red LED OFF

4. Vehicle Entry Detection Algorithm

The IR sensor (ir4) detects vehicle arrival at the gate.

- If vehicle detected → system checks slot availability
- If no vehicle → system remains idle

5. Gate Control Algorithm (Servo Motor)

When a vehicle arrives:

- If any slot is empty → Servo opens gate (110°), then closes
- If all slots are full → Gate remains closed

6. Buzzer Alert Algorithm

If parking is full and a vehicle tries to enter:

- Buzzer is activated to alert the driver

7. IoT Data Transmission Algorithm

The system sends parking data to the server:

- Slot status (Full/Empty) is converted into strings
- Data is sent using HTTP GET request via Wi-Fi
- Server stores and updates real-time parking status

8. Wi-Fi Connectivity Algorithm

The ESP32 connects to Wi-Fi using SSID and password:

- Continuously checks connection status
- Proceeds only after successful connection

9. Continuous Monitoring Algorithm

The loop() function runs continuously:

- Reads sensor values

- Updates display and LEDs
- Controls gate and sends IoT data

Modules Used in Project

The IoT-Based Vehicle Parking Management System is divided into several modules, each responsible for a specific function to ensure smooth operation of the system.

The **Sensor Module** is used to detect the presence of vehicles in parking slots. IR sensors continuously monitor each slot and send signals to the microcontroller indicating whether the slot is occupied or empty.

The **Processing Module (ESP32 Microcontroller)** acts as the brain of the system. It receives data from sensors, processes it, and makes decisions such as updating slot status, controlling the gate, and sending data to the server.

The **Display Module** includes an LCD display that shows real-time parking availability (Full/Empty) for each slot, helping users quickly identify free spaces.

The **Gate Control Module** uses a servo motor to automatically open and close the entry gate based on parking availability and vehicle detection.

The **Communication Module (Wi-Fi)** enables the system to connect to the internet and send parking data to a remote server using HTTP protocol for real-time monitoring.

The **Alert Module** consists of a buzzer and LED indicators. The buzzer alerts when parking is full, and LEDs indicate whether slots are available (green) or full (red).

The **IoT Cloud Module** is responsible for storing and updating parking data on a server, allowing users to access the information through a web or mobile application.

4.3 SAMPLE CODE:

4.3.1 Project Setup and Configuration

The setup of the IoT-Based Vehicle Parking Management System begins with assembling the hardware components. Sensors such as ultrasonic or IR sensors are installed in each parking slot and connected to the microcontroller (Arduino/NodeMCU). The servo motor is connected to control the entry and exit gate, while the LCD display is used to show available parking slots. All components are powered using a suitable power supply and connected using jumper wires or a PCB. <!DOCTYPE html>

```
#include <ESP32Servo.h>
Servo myservo;  // create servo object to control a servo

#include <LiquidCrystal.h>
#include <stdio.h>
//LiquidCrystal lcd(6, 7, 5, 4, 3, 2);  //rs,en,d4,d5,d6,d7
LiquidCrystal lcd(13, 12, 14, 27, 26, 25);


#include <WiFi.h>
#include <HTTPClient.h>

HTTPClient http;

const char* ssid = "iotserver";
const char* password = "iotserver123";

char gchr='x';


int httpResponseCode;
String servername =
"http://projectsfactoryserver.in/storedata.php?name=";
String accountname = "iot1684";
String field1 = "&s1=";
String field2 = "&s2=";
String field3 = "&s3=";
String field4 = "&s4=";
String field5 = "&s5=";
String field6 = "&s6=";
String field7 = "&s7=";
String field8 = "&s8=";
String field9 = "&s9=";
String payload="";

String servername_read =
"http://projectsfactoryserver.in/last.php?name=";
String accountname_read = "iot1067L";

char ir1s='e',ir2s='e',ir3s='e';
String slot1_string="";
String slot2_string="";
String slot3_string="";

int amount1=1000,balance1=0;
int amount2=1000,balance2=0;

char rfid_string[13];
int cardv=0;
int ii=0;

int sti=0;
```

```
String inputString = "";         // a string to hold incoming data
boolean stringComplete = false;  // whether the string is complete


char rcv,pastnumber[11];

int ir1   = 21;
int ir2   = 19;
int ir3   = 18;

int buzzer = 16;

int ir4 = 5;

int red_led   = 23;
int green_led = 22;

void iot_send()
{
      lcd.setCursor(15,0);lcd.print("U");
      http.begin(servername + accountname + field1 + slot1_string +
field2 + slot2_string + field3 + slot3_string);
      httpResponseCode = http.GET();
      if(httpResponseCode>0)
        {
         payload="";
         //Serial.print("HTTP Response code: ");
         //Serial.println(httpResponseCode);
         payload = http.getString();
         //Serial.println(payload);
        }
      else
        {
         ;
         //Serial.print("Error code: ");
         //Serial.println(httpResponseCode);
        }
      delay(8000);
      lcd.setCursor(15,0);lcd.print(" ");
}

void beep()
{
  digitalWrite(buzzer, LOW);delay(2000); digitalWrite(buzzer,
HIGH);delay(200);
}
void okcheck0()
{
  unsigned char rcr;
  do{
      rcr = Serial.read();
    }while(rcr != 'K');
}
void setup()
{
  Serial.begin(9600);

  myservo.attach(17);
  myservo.write(10);

  pinMode(ir1, INPUT);pinMode(ir2, INPUT);pinMode(ir3, INPUT);
  pinMode(ir4, INPUT);
  pinMode(buzzer, OUTPUT);
```

```
  pinMode(red_led, OUTPUT);pinMode(green_led, OUTPUT);

  digitalWrite(buzzer, HIGH);
  digitalWrite(red_led, HIGH);digitalWrite(green_led, HIGH);

  lcd.begin(16, 2);lcd.cursor();
  lcd.print("   IOT Vehicle");
  lcd.setCursor(0,1);
  lcd.print("Parking Manager");
  delay(2000);

  WiFi.begin(ssid, password);
  while(WiFi.status() != WL_CONNECTED)
       {
           delay(500);
       }
  delay(3000);

 lcd.clear();
 lcd.print("S1:");//3-4,0
 lcd.setCursor(5,0);
 lcd.print("S2:");//8-9,0
 lcd.setCursor(10,0);
 lcd.print("S3:");//13-14,0
 delay(500);

 // serialEvent();
}

int cntlmk=0;
//char ir1s='e',ir2s='e';

int s1v=0,s2v=0;

void loop()
{
       slot1_string = "";
       if(digitalRead(ir1) == LOW)
         {
          lcd.setCursor(3,0);lcd.print("Fl");
          ir1s='f';
          slot1_string="Full";
         }
       if(digitalRead(ir1) == HIGH)
         {
          lcd.setCursor(3,0);lcd.print("Em");
          ir1s='e';
          slot1_string="Empty";
         }

       slot2_string="";
       if(digitalRead(ir2) == LOW)
         {
          lcd.setCursor(8,0);lcd.print("Fl");
          ir2s='f';
          slot2_string="Full";
         }
       if(digitalRead(ir2) == HIGH)
         {
          lcd.setCursor(8,0);lcd.print("Em");
          ir2s='e';
          slot2_string="Empty";
         }
```

```
       slot3_string="";
       if(digitalRead(ir3) == LOW)
         {
          lcd.setCursor(13,0);lcd.print("Fl");
          ir3s='f';
          slot3_string="Full";
         }
       if(digitalRead(ir3) == HIGH)
         {
          lcd.setCursor(13,0);lcd.print("Em");
          ir3s='e';
          slot3_string="Empty";
         }

       if(digitalRead(ir1) == LOW && digitalRead(ir2) == LOW &&
digitalRead(ir3) == LOW)
         {
          digitalWrite(red_led, LOW);digitalWrite(green_led, HIGH);
         }
       else
         {
          digitalWrite(red_led, HIGH);digitalWrite(green_led, LOW);
         }

     if(digitalRead(ir4) == LOW)
       {
        if(ir1s == 'e' || ir2s == 'e' || ir3s == 'e')
          {
           myservo.write(110);
           delay(2000);
           myservo.write(10);delay(1000);

           iot_send();
          }
        else
          {
            beep();
            iot_send();
          }
       }

    delay(500);
}


void convertl(unsigned int value)
{
  unsigned int a,b,c,d,e,f,g,h;

      a=value/10000;
      b=value%10000;
      c=b/1000;
      d=b%1000;
      e=d/100;
      f=d%100;
      g=f/10;
      h=f%10;


      a=a|0x30;
      c=c|0x30;
      e=e|0x30;
```

```
        g=g|0x30;
        h=h|0x30;


    //lcd.write(a);
    //lcd.write(c);
    lcd.write(e);
    lcd.write(g);
    lcd.write(h);
}
```

SYSTEM ENVIRONMENT

The system environment of the IoT-Based Vehicle Parking Management System is built by combining embedded hardware components with IoT-based communication to achieve a smart and automated parking solution. The central unit of the system is the ESP32 microcontroller, which controls all operations and processes data received from sensors. It is programmed using Embedded C/C++ in the Arduino IDE, where the logic for parking detection, gate control, and data transmission is implemented.

The hardware environment consists of multiple IR sensors placed in parking slots to detect vehicle presence, and an additional IR sensor at the entry point to detect incoming vehicles. A servo motor is used to automatically open and close the parking gate based on slot availability. An LCD display is used to show real-time status of parking slots (Full/Empty), while LEDs indicate overall availability (green for available, red for full). A buzzer is included to alert users when the parking area is full.

The ESP32 uses its built-in Wi-Fi module to connect to a wireless network using predefined SSID and password. Once connected, it sends real-time data to a remote server using HTTP protocol. The data includes the status of each parking slot, which is updated continuously. This allows users to monitor parking availability remotely through a web interface or mobile application.

The software environment includes Arduino IDE for development and required libraries such as WiFi.h for connectivity, HTTPClient.h for server communication, LiquidCrystal.h for LCD display control, and ESP32Servo.h for servo motor operation. These libraries simplify the integration of different hardware components.

The system is designed to work in a continuous loop, where sensor data is constantly monitored, processed, and updated. It supports real-time operation with minimal delay and ensures reliable communication between hardware and server. The environment is also scalable, meaning additional parking slots and sensors can be added easily without major changes to the system.

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

5.1 System Power ON Screen

When the system is powered ON, the LCD display shows the project title “IoT Vehicle Parking Manager”. This confirms that all hardware components are initialized successfully.

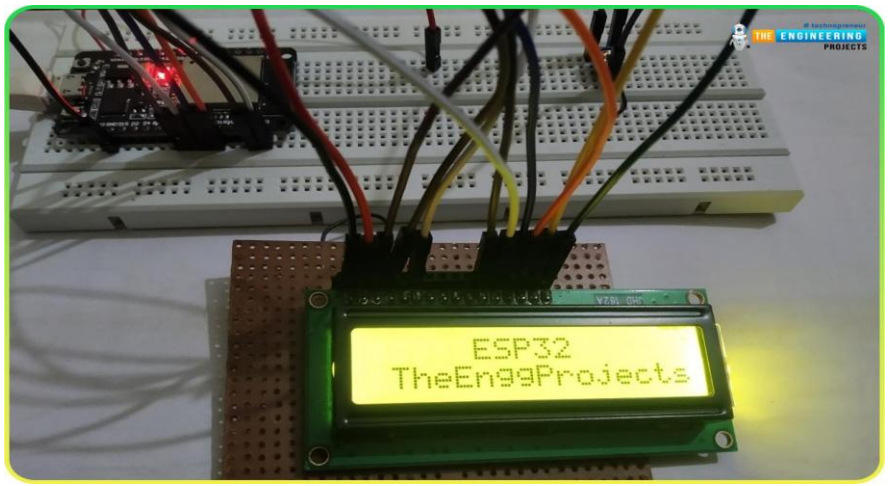

Figure 5.1 : ESP32 with LCD Display Setup for IoT Vehicle Parking System

5.2 Wi-Fi Connection Setup

After initialization, the ESP32 connects to the Wi-Fi network using SSID and password. Once connected, the system is ready for IoT communication.

Figure 5.2: ESP32 Interfaced with 16×2 LCD Display Showing Output.

5.3 Parking Slot Status Display

The LCD shows slot status like S1, S2, S3 with “Em” (Empty) or “Fl” (Full), helping users identify free spaces**.**

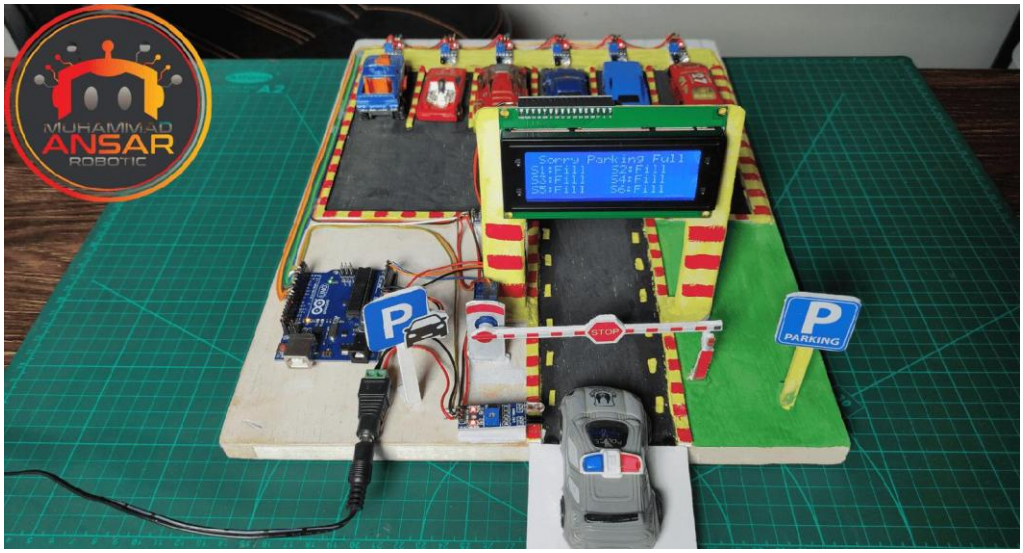

Figure 5.3: Parking Slot Status Display Using LCD

5.4 Vehicle Detection at Entry

When a vehicle arrives, the IR sensor detects it and sends a signal to the controller to check slot availability.

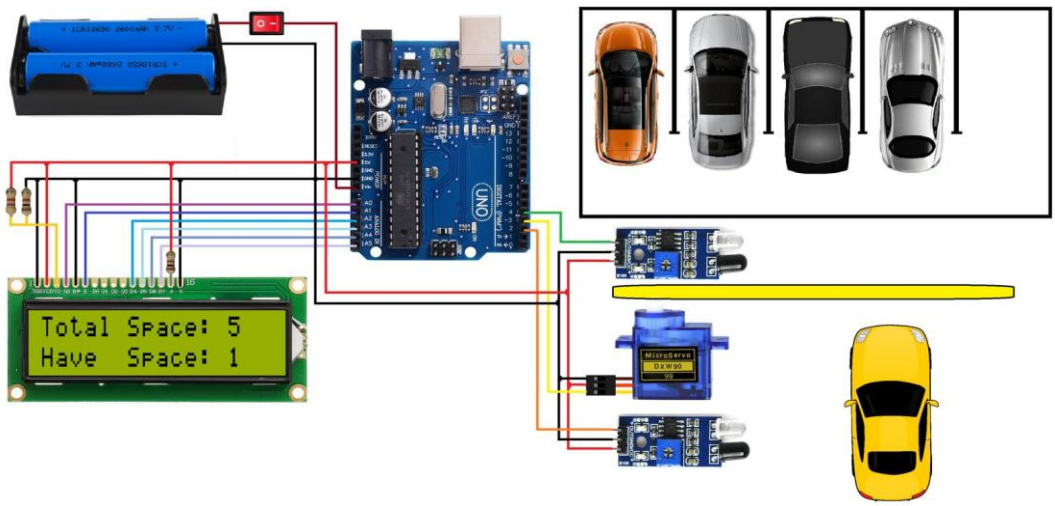

Figure 5.4:Vehicle Detection at Entry Point Using IR Sensor in Smart Parking System

5.5 Gate Opening Mechanism

If parking is available, the servo motor opens the gate automatically and closes it after a delay.

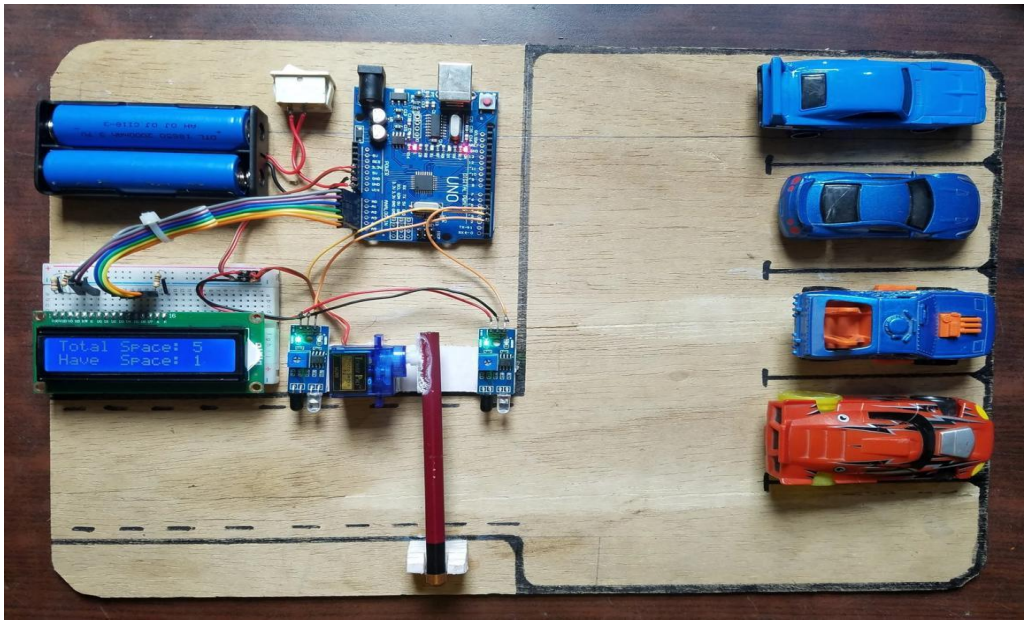

Figure 5.5:Automatic Gate Opening Mechanism Using Servo Motor in Smart Parking System

5.6 Parking Full

If all slots are full, the buzzer alerts the user and the gate does not open.

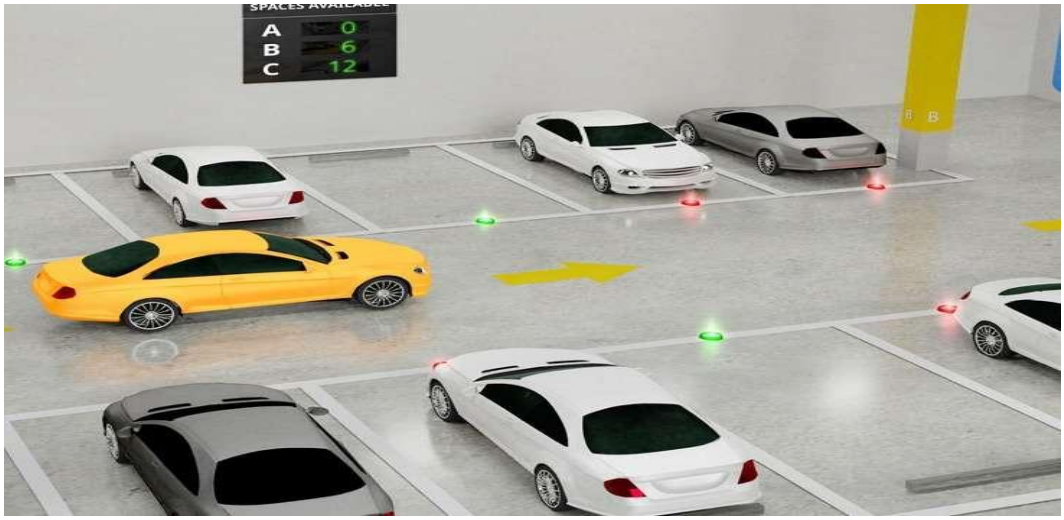

Figure 5.6: Smart Parking System Display Indicating Parking Full Status

5.7 LED Indicator

LED indicators show status: Green for available slots and Red for full parking.

Figure 5.7: LED Indicator Showing Parking Slot Availability in Smart Parking System.

6. VALIDATION

6. VALIDATION

6.1 INTRODUCTION

Validation is an important phase in the IoT-Based Vehicle Parking Management System to ensure that the system works correctly and meets the intended requirements. It involves testing the complete system under different conditions to verify the accuracy, reliability, and performance of both hardware and software components.

In this project, validation is carried out by checking the proper functioning of sensors, microcontroller, communication module, and output devices. The system is tested to ensure that parking slots are correctly detected as “Full” or “Empty”, the gate opens only when space is available, and alerts are generated when the parking area is full.

Additionally, the IoT functionality is validated by confirming that data is successfully transmitted to the server and updated in real time. The overall system is observed for continuous operation, quick response, and error-free performance.

6.2 TEST CASES

6.2.1 Data Upload / IoT Data Transmission

Test Case	**Data Upload to Server**
Input	Sensor Data + Wi-Fi Connection
Process	ESP32 sends data using HTTP
Expected Output	Data updated on server
Actual Output	Data successfully uploaded
Result	PASS

6.2.2 Sensor Detection

Test Case	**Slot Detection**
Input	IR Sensor Signal
Process	Detect Full / Empty
Expected Output	Correct slot status
Actual Output	Slot displayed correctly
Result	PASS

6.2.3 Gate Control

Test Case	**Gate Operation**
Input	Vehicle at Entry
Process	Servo opens/closes gate
Expected Output	Gate opens if slot available
Actual Output	Gate works correctly
Result	PASS

6.2.4 LCD Display

Test Case	**Display Status**
Input	Sensor Data
Process	Update LCD
Expected Output	Show Full / Empty
Actual Output	Display works correctly
Result	PASS

7. CONCLUSION & FUTURE WORK

7. CONCLUSION & FUTURE WORK

The IoT Based Vehicle Parking Manager offers an efficient, reliable, and intelligent solution to the growing parking challenges in urban environments. By leveraging IoT technologies such as sensors, microcontrollers, wireless communication, and cloud computing, the system enables real-time monitoring and management of parking spaces. This significantly reduces the time spent by drivers searching for parking slots, thereby lowering traffic congestion, fuel consumption, and air pollution.

The automation of parking operations minimizes human intervention, reduces errors, and enhances overall system accuracy and efficiency. Additionally, the availability of real-time data allows authorities to analyze parking patterns, optimize space utilization, and plan future infrastructure improvements. The system is scalable, cost-effective, and adaptable to various environments such as shopping malls, offices, hospitals, and smart cities.

In conclusion, the IoT Based Vehicle Parking Manager not only improves user convenience but also contributes to sustainable urban development. With further enhancements such as AI-based prediction, smart navigation, and seamless payment integration, this system can play a vital role in shaping the future of intelligent transportation and smart city ecosystems.

In the future, the system can be enhanced by adding advanced features such as mobile app integration for booking parking slots in advance. RFID or number plate recognition can be implemented for better security and automatic user identification. Online payment systems can be integrated for seamless transactions. The system can also be expanded to support a larger number of parking slots and integrated with smart city infrastructure. Additionally, machine learning techniques can be used to predict parking demand and optimize space utilization more efficiently.

8. REFERENCE

8. REFERENCES

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8.2 GITHUB LINK

The complete source code for this project is available at the following GitHub repository:

https://github.com/Manojkumar128/Iot-based-vehicle-parking-management-system.git